

POWER BI – WEEK 1 PROJECT REPORT

1. Introduction

The Superstore Sales project focuses on preparing retail sales data for business intelligence and future dashboard development. The raw Superstore dataset contains information about customers, products, orders, sales, quantities, discounts, profits and regions.

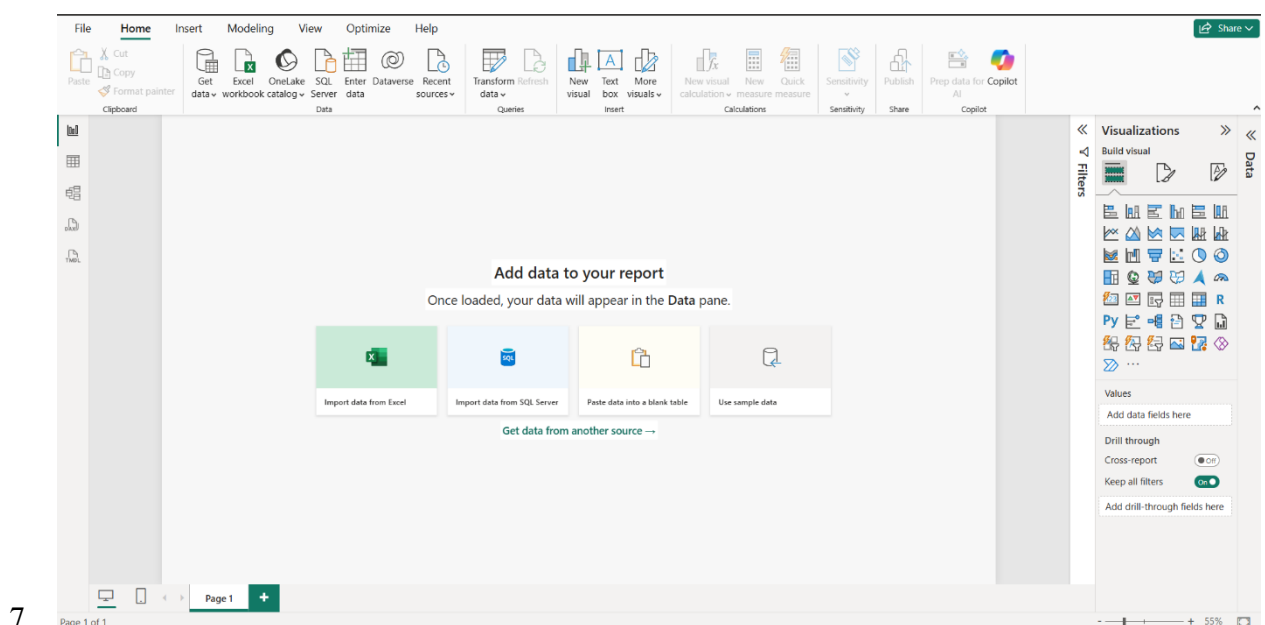
In Week 1, Power BI and Power Query were used to import, inspect, clean and transform the raw dataset. The cleaned dataset will be used for data modeling and dashboard development in the following week.

Tools Used

- Microsoft Power BI Desktop
- Power Query Editor
- Superstore Sales Dataset

Objectives

1. Import the Superstore dataset into Power BI.
2. Inspect the dataset and its attributes.
3. Verify and correct data types.
4. Clean and standardize categorical columns.
5. Create date-related columns.
6. Load the cleaned dataset into Power BI.



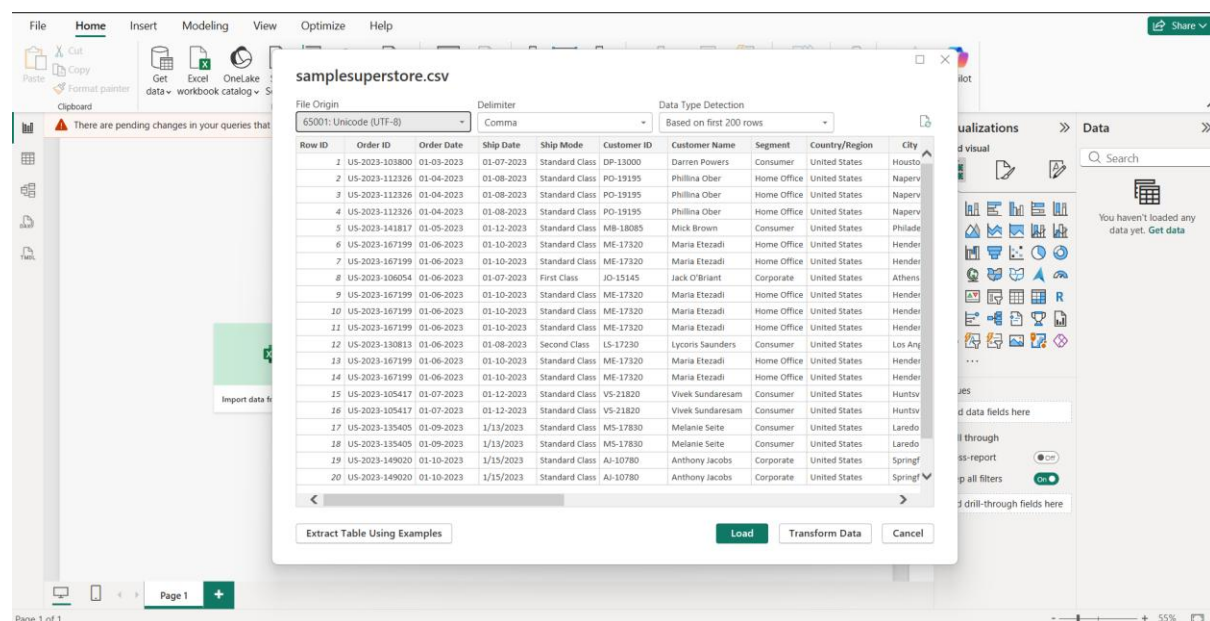
2. Data Import

The Superstore Sales dataset was imported into Power BI Desktop using the **Text/CSV** option.

Steps

1. Opened Power BI Desktop.
2. Selected **Get Data**.
3. Selected **Text/CSV**.
4. Selected the Superstore dataset.
5. Previewed the data.
6. Selected **Transform Data** to open Power Query Editor.

Power Query was selected because the dataset needed to be cleaned and transformed before analysis.



3. Data Inspection

The imported dataset was inspected in Power Query Editor. The column names, data values, data types and missing values were checked.

Important fields such as Order Date, Sales, Quantity, Profit, Category, Sub-Category and Segment were verified.

3.1 Data Type Correction

The data types were corrected according to the nature of the data.

For example:

- Order Date → Date
- Sales → Decimal Number
- Quantity → Whole Number
- Profit → Decimal Number
- Category → Text
- Segment → Text

HomeTransformAdd ColumnViewTools

Queries1

Table.TransformColumnTypes(*Promoted Headers*,([Row ID", Int64.Type), ("Order ID", type text), ("Order Date", type text), ("Ship Date", type text), ("Ship Mode", type text), ("Customer ID", type text), ("Customer Name", type text))

samplesuperstore

Row ID

Order ID

Order Date

Ship Date

Ship Mode

Customer ID

Customer Name

Valid

100%

Valid

100%

Valid

100%

Valid

100%

Valid

100%

Error

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Error

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Error

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Error

0%

Error

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Empty

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Empty

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1

2

US-2023-101800

01-03-2023

01-07-2023

Standard Class

DP-13000

Darren Power

2

2

US-2023-112326

01-04-2023

01-08-2023

Standard Class

PO-19195

Phillina Ober

3

3

US-2023-112326

01-04-2023

01-08-2023

Standard Class

PO-19195

Phillina Ober

4

4

US-2023-112326

01-04-2023

01-08-2023

Standard Class

PO-19195

Phillina Ober

5

5

US-2023-141817

01-05-2023

01-12-2023

Standard Class

MB-18085

Nick Brown

6

6

US-2023-167199

01-06-2023

01-10-2023

Standard Class

ME-17320

Maria Etezad

7

7

US-2023-167199

01-06-2023

01-10-2023

Standard Class

ME-17320

Maria Etezad

8

8

US-2023-100054

01-06-2023

01-07-2023

First Class

JO-15145

Jack O'Brian

9

9

US-2023-167199

01-06-2023

01-10-2023

Standard Class

ME-17320

Maria Etezad

10

10

US-2023-167199

01-06-2023

01-10-2023

Standard Class

ME-17320

Maria Etezad

11

11

US-2023-167199

01-06-2023

01-10-2023

Standard Class

ME-17320

Maria Etezad

12

12

US-2023-130813

01-06-2023

01-08-2023

Second Class

LS-17230

Lycoris Saund

13

13

US-2023-167199

01-06-2023

01-10-2023

Standard Class

ME-17320

Maria Etezad

14

14

US-2023-167199

01-06-2023

01-10-2023

Standard Class

ME-17320

Maria Etezad

15

15

US-2023-105417

01-07-2023

01-12-2023

Standard Class

VS-21820

Vivek Sundan

16

16

US-2023-105417

01-07-2023

01-12-2023

Standard Class

VS-21820

Vivek Sundan

17

17

US-2023-135405

01-09-2023

1/13/2023

Standard Class

MS-17830

Melanie Sette

18

18

US-2023-135405

01-09-2023

1/13/2023

Standard Class

MS-17830

Melanie Sette

19

19

US-2023-149020

01-10-2023

1/15/2023

Standard Class

AJ-10780

Anthony Jaco

20

20

US-2023-149020

01-10-2023

1/15/2023

Standard Class

AJ-10780

Anthony Jaco

21

21

US-2023-130092

01-11-2023

1/14/2023

First Class

SV-20865

Seth Vernon

22

22

US-2023-157147

1/13/2023

1/18/2023

Standard Class

BD-11605

Brian Dahlen

23

23

US-2023-109232

1/13/2023

1/16/2023

Second Class

ND-18370

Natalie DeCh

24

24

US-2023-157147

1/13/2023

1/18/2023

Standard Class

BD-11605

Brian Dahlen

25

25

US-2023-118192

1/13/2023

1/18/2023

Standard Class

MM-17920

Michael Moo

26

26

US-2023-162775

1/13/2023

1/15/2023

Second Class

CS-12250

Chris Selesnic

27

27

US-2023-162775

1/13/2023

1/15/2023

Second Class

CS-12250

Chris Selesnic

28

28

US-2023-162775

1/13/2023

1/15/2023

Second Class

CS-12250

Chris Selesnic

29

29

US-2023-118192

1/13/2023

1/18/2023

Standard Class

MM-17920

Michael Moo

21 COLUMNS, 999+ ROWS

Column profiling based on top 1000 rows

Query Settings

PROPERTIES

Name

samplesuperstore

All Properties

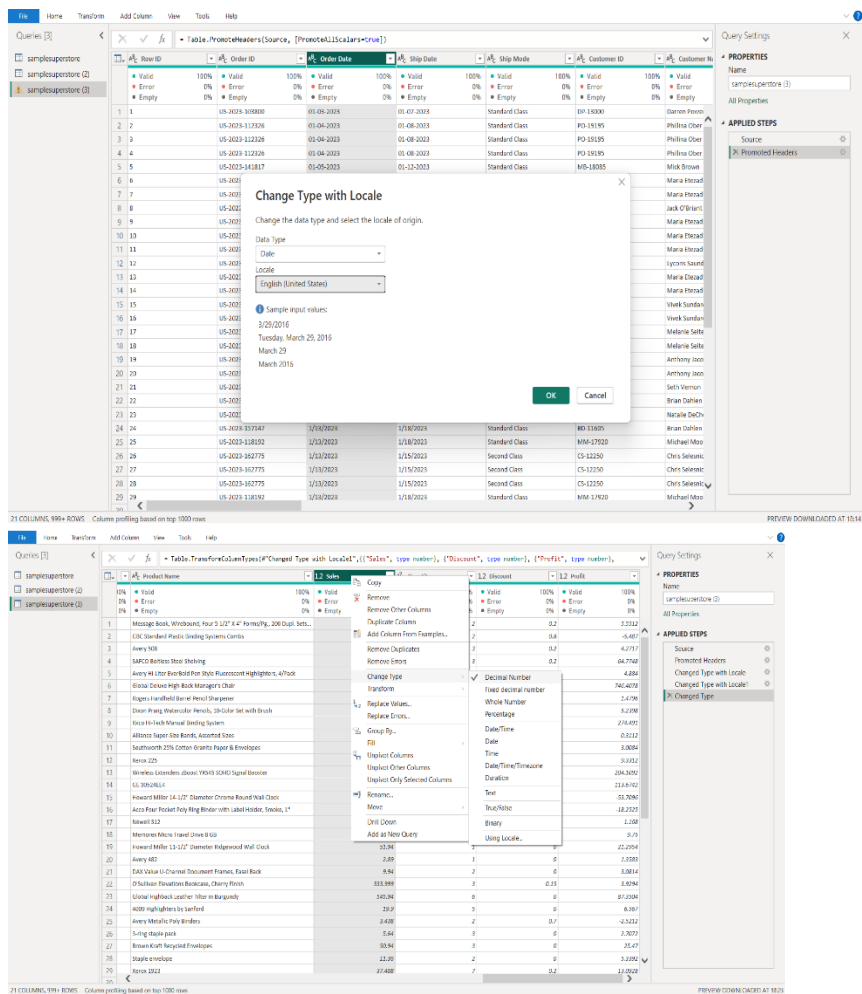
APPLIED STEPS

Source

Promoted Headers

Changed Type

PREVIEW DOWNLOADED AT 15:00



4. Text Data Cleaning

Categorical columns such as **Category**, **Sub-Category** and **Segment** were standardized.

The **Trim** transformation was used to remove unnecessary spaces from text values. The **Clean** transformation was used to remove unwanted non-printable characters.

Steps

1. Selected the categorical columns.
2. Selected **Transform**.
3. Selected **Format**.
4. Applied **Trim**.
5. Applied **Clean** where required.
6. Verified the resulting values.

This helped maintain consistent categorical values and avoid duplicate categories caused by unwanted spaces or characters.

Order ID	Product Name	Category	Sub-Category	Quantity	Price	Discount	Total
1	Office Supplies	Office Supplies	Office Supplies	1	1.00	0.00	1.00
2	Office Supplies	Office Supplies	Office Supplies	1	1.00	0.00	1.00
3	Office Supplies	Office Supplies	Office Supplies	1	1.00	0.00	1.00
4	Office Supplies	Office Supplies	Office Supplies	1	1.00	0.00	1.00
5	Office Supplies	Office Supplies	Office Supplies	1	1.00	0.00	1.00
6	Furniture	Furniture	Furniture	1	1.00	0.00	1.00
7	Office Supplies	Office Supplies	Office Supplies	1	1.00	0.00	1.00
8	Office Supplies	Office Supplies	Office Supplies	1	1.00	0.00	1.00
9	Office Supplies	Office Supplies	Office Supplies	1	1.00	0.00	1.00
10	Office Supplies	Office Supplies	Office Supplies	1	1.00	0.00	1.00
11	Office Supplies	Office Supplies	Office Supplies	1	1.00	0.00	1.00
12	Office Supplies	Office Supplies	Office Supplies	1	1.00	0.00	1.00
13	Technology	Technology	Technology	1	1.00	0.00	1.00
14	Technology	Technology	Technology	1	1.00	0.00	1.00
15	Furniture	Furniture	Furniture	1	1.00	0.00	1.00
16	Office Supplies	Office Supplies	Office Supplies	1	1.00	0.00	1.00
17	Office Supplies	Office Supplies	Office Supplies	1	1.00	0.00	1.00
18	Technology	Technology	Technology	1	1.00	0.00	1.00
19	Furniture	Furniture	Furniture	1	1.00	0.00	1.00
20	Office Supplies	Office Supplies	Office Supplies	1	1.00	0.00	1.00
21	Furniture	Furniture	Furniture	1	1.00	0.00	1.00
22	Furniture	Furniture	Furniture	1	1.00	0.00	1.00
23	Furniture	Furniture	Furniture	1	1.00	0.00	1.00
24	Office Supplies	Office Supplies	Office Supplies	1	1.00	0.00	1.00
25	Office Supplies	Office Supplies	Office Supplies	1	1.00	0.00	1.00
26	Office Supplies	Office Supplies	Office Supplies	1	1.00	0.00	1.00
27	Office Supplies	Office Supplies	Office Supplies	1	1.00	0.00	1.00
28	Office Supplies	Office Supplies	Office Supplies	1	1.00	0.00	1.00
29	Office Supplies	Office Supplies	Office Supplies	1	1.00	0.00	1.00
30	Office Supplies	Office Supplies	Office Supplies	1	1.00	0.00	1.00

5. Creating Date Attributes

The Order Date column was used to create additional temporal attributes.

The following columns were created:

New Column Purpose

Order Year Analyze yearly sales

Order Quarter Analyze quarterly performance

Order Month Analyze monthly performance

Order Day Analyze daily information

Day of Week Analyze weekday patterns

These columns will be useful for time-based analysis and visualization in the dashboard.

File Home Transform Add Column View Tools Help

Queries [3] Table.RenameColumns(#"Inserted Week of Year",{"Week of Year", "Order Week of Year"})

	1.2 Profit	1.3 Order Year	1.4 Order Quarter	1.5 Order Month Name	1.6 Order Day	1.7 Order Week of Year
1	0.2	5.5512	2023	1 January	3	1
2	0.8	-5.487	2023	1 January	4	1
3	0.2	4.2717	2023	1 January	4	1
4	0.2	-64.7748	2023	1 January	4	1
5	0.2	4.884	2023	1 January	5	1
6	0	746.4078	2023	1 January	6	1
7	0	1.4796	2023	1 January	6	1
8	0	5.2398	2023	1 January	6	1
9	0	274.491	2023	1 January	6	1
10	0	0.3112	2023	1 January	6	1
11	0	3.0084	2023	1 January	6	1
12	0	9.3312	2023	1 January	6	1
13	0	204.1092	2023	1 January	6	1
14	0	113.6742	2023	1 January	6	1
15	0.6	-53.7096	2023	1 January	7	1
16	0.8	-18.2525	2023	1 January	7	1
17	0.2	1.168	2023	1 January	9	2
18	0.2	9.75	2023	1 January	9	2
19	0	21.2954	2023	1 January	10	2
20	0	1.3583	2023	1 January	10	2
21	0	3.0814	2023	1 January	11	2
22	0.15	3.9294	2023	1 January	13	2
23	0	87.3504	2023	1 January	13	2
24	0	6.567	2023	1 January	13	2
25	0.7	-2.5212	2023	1 January	13	2
26	0	2.7072	2023	1 January	13	2
27	0	25.47	2023	1 January	13	2
28	0	5.3392	2023	1 January	13	2
29	0.2	13.0928	2023	1 January	13	2

26 COLUMNS, 999+ ROWS Column profiling based on top 1000 rows

Query Settings

NAME
samplesuperstore (3)

APPLIED STEPS

- Source
- Promoted Headers
- Changed Type with Locale
- Changed Type with Locale1
- Trimmed Text
- Cleaned Text
- Inserted Year
- Renamed Columns
- Inserted Quarter
- Renamed Columns1
- Inserted Month Name
- Renamed Columns2
- Inserted Day
- Renamed Columns3
- Inserted Week of Year
- Renamed Columns4

PREVIEW DOWNLOADED AT 18:42

6. Applying Transformations

After completing all required cleaning and transformation activities, **Close & Apply** was selected in Power Query.

Power BI applied the transformations and loaded the cleaned dataset into the data model.

File

Home

Transform

Add Column

View

Tools

Help

Close & Apply

New Source

Recent Sources

Enter Data

Data source settings

Manage Parameters

Export query results

Refresh Preview

Advanced Editor

Properties

Choose Columns

Remove Columns

Keep Rows

Remove Rows

Split Column

Group By

Data Type: Whole Number

Use First Row as Headers

Use First Row as Headers

Replace Values

Merge Queries

Append Queries

Combine Queries

Combine Files

Close

New Query

Data Sources

Parameters

Output Data

Query

Manage Columns

Reduce Rows

Sort

Transform

Combine

samplesuperstore (3)

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US-2023-112326

US-2023-112326

US-2023-112326

US-2023-141817

US-2023-187199

US-2023-187199

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US-2023-187199

US-2023-130813

US-2023-187199

US-2023-187199

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US-2023-105417

US-2023-135405

US-2023-135405

US-2023-149020

US-2023-149020

US-2023-130092

US-2023-157147

US-2023-102932

US-2023-157147

US-2023-118192

US-2023-162775

US-2023-162775

US-2023-162775

US-2023-118192

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Second Class

Standard Class

DP-13000

PO-19195

PO-19195

PO-19195

MB-18085

ME-17320

ME-17320

JO-15145

ME-17320

ME-17320

ME-17320

LS-17230

ME-17320

ME-17320

V5-21820

V5-21820

MS-17830

MS-17830

AJ-10780

AJ-10780

SV-20365

BD-11605

ND-18370

BD-11605

MM-17920

CS-12250

CS-12250

CS-12250

MM-17920

Darren Power

Phillina Ober

Phillina Ober

Phillina Ober

Mick Brown

Maria Etead

Maria Etead

Jack O'Brian

Maria Etead

Maria Etead

Maria Etead

Lycoris Saund

Maria Etead

Maria Etead

Vivek Sundan

Vivek Sundan

Melanie Seite

Melanie Seite

Anthony Jaco

Anthony Jaco

Seth Vernon

Brian Dahlin

Natalie DeCh

Brian Dahlin

Michael Moo

Chris Selesnic

Chris Selesnic

Chris Selesnic

Michael Moo

APPLIED STEPS

Source

Promoted Headers

Changed Type

7. Final Dataset

The final dataset was checked in Power BI Data View. The cleaned categorical columns, corrected data types and newly created date columns were verified.

FileHomeHelpTable tools

Namesamplesuperstore (3)

Manage relationships

New measure

Quick measure

New column

New table

Mark as date table

StructureRelationshipsCalculationsCalendars

Row ID

Order ID

Order Date

Ship Date

Ship Mode

Customer ID

Customer Name

Segment

Country/Region

City

State/Province

Postal Code

Region

12

US-2023-130813

06 January 2023

08 January 2023

Second Class

LS-17230

Lycoris Saunders

Consumer

United States

Los Angeles

California

90049

West

48

US-2023-148614

20 January 2023

25 January 2023

Standard Class

MV-17485

Mark Van Huff

Consumer

United States

Los Angeles

California

90049

West

57

US-2023-148614

20 January 2023

25 January 2023

Standard Class

MV-17485

Mark Van Huff

Consumer

United States

Los Angeles

California

90049

West

122

US-2023-164903

20 February 2023

24 February 2023

Standard Class

SR-20740

Steven Roelle

Home Office

United States

Los Angeles

California

90049

West

324

US-2023-102652

06 April 2023

12 April 2023

Standard Class

AY-10555

Andy Yotov

Corporate

United States

Los Angeles

California

90049

West

325

US-2023-102652

06 April 2023

12 April 2023

Standard Class

AY-10555

Andy Yotov

Corporate

United States

Los Angeles

California

90049

West

331

US-2023-102652

06 April 2023

12 April 2023

Standard Class

AY-10555

Andy Yotov

Corporate

United States

Los Angeles

California

90049

West

332

US-2023-102652

06 April 2023

12 April 2023

Standard Class

AY-10555

Andy Yotov

Corporate

United States

Los Angeles

California

90049

West

369

US-2023-120740

15 April 2023

15 April 2023

Same Day

PS-18970

Paul Stevenson

Home Office

United States

Los Angeles

California

90049

West

370

US-2023-120740

15 April 2023

15 April 2023

Same Day

PS-18970

Paul Stevenson

Home Office

United States

Los Angeles

California

90049

West

373

US-2023-110849

18 April 2023

23 April 2023

Standard Class

JL-15835

John Lee

Consumer

United States

Los Angeles

California

90049

West

375

US-2023-110849

18 April 2023

23 April 2023

Standard Class

JL-15835

John Lee

Consumer

United States

Los Angeles

California

90049

West

377

US-2023-110849

18 April 2023

23 April 2023

Standard Class

JL-15835

John Lee

Consumer

United States

Los Angeles

California

90049

West

723

US-2023-114125

09 July 2023

13 July 2023

Standard Class

GH-14410

Gary Hansen

Home Office

United States

Los Angeles

California

90049

West

727

US-2023-114125

09 July 2023

13 July 2023

Standard Class

GH-14410

Gary Hansen

Home Office

United States

Los Angeles

California

90049

West

1409

US-2023-101931

28 October 2023

31 October 2023

First Class

TS-21370

Todd Sumrall

Corporate

United States

Los Angeles

California

90049

West

1410

US-2023-101931

28 October 2023

31 October 2023

First Class

TS-21370

Todd Sumrall

Corporate

United States

Los Angeles

California

90049

West

1411

US-2023-101931

28 October 2023

31 October 2023

First Class

TS-21370

Todd Sumrall

Corporate

United States

Los Angeles

California

90049

West

1422

US-2023-106439

31 October 2023

04 November 2023

Standard Class

GG-14650

Greg Guthrie

Corporate

United States

Los Angeles

California

90049

West

1427

US-2023-106439

31 October 2023

04 November 2023

Standard Class

GG-14650

Greg Guthrie

Corporate

United States

Los Angeles

California

90049

West

1428

US-2023-106439

31 October 2023

04 November 2023

Standard Class

GG-14650

Greg Guthrie

Corporate

United States

Los Angeles

California

90049

West

1430

US-2023-106439

31 October 2023

04 November 2023

Standard Class

GG-14650

Greg Guthrie

Corporate

United States

Los Angeles

California

90049

West

1431

US-2023-106439

31 October 2023

04 November 2023

Standard Class

GG-14650

Greg Guthrie

Corporate

United States

Los Angeles

California

90049

West

1432

US-2023-106439

31 October 2023

04 November 2023

Standard Class

GG-14650

Greg Guthrie

Corporate

United States

Los Angeles

California

90049

West

1446

US-2023-168305

01 November 2023

08 November 2023

Standard Class

PL-18925

Paul Lucas

Home Office

United States

Los Angeles

California

90049

West

1449

US-2023-168305

01 November 2023

08 November 2023

Standard Class

PL-18925

Paul Lucas

Home Office

United States

Los Angeles

California

90049

West

1468

US-2023-153619

03 November 2023

03 November 2023

Same Day

TC-21535

Tracy Collins

Home Office

United States

Los Angeles

California

90049

West

Search

samplesuperstore

samplesuperstore (2)

samplesuperstore (3)

Table: samplesuperstore (3) [10,194 rows]

Table: samplesuperstore (3) (10,194 rows)